SIMATS ENGINEERING

Assessment Tool 2: Scenario-Based — Suggested Solutions

Course: Computer Vision (ITA05) | CO2: Apply image enhancement and segmentation techniques for extracting meaningful regions from images (BTL 3)

Note: This is a model/reference answer set to help you understand the reasoning expected for each part (a)-(e). Rewrite it in your own words and add your registration details before signing the Code of Conduct declaration, since the assessment requires original, individually authored work.

1. Salt-and-Pepper Noise (Scanned Document)

(a) Scenario & Characteristics: A scanned document affected by salt-and-pepper noise shows scattered pixels that are either pure white (255) or pure black (0), randomly distributed across the image. This impulsive noise typically results from faulty scanner sensors, dust on the glass bed, dead pixels in the imaging array, or bit errors during transmission/storage. Unlike Gaussian noise, it does not affect every pixel — only a random subset is corrupted, but those pixels are pushed to extreme intensity values.

(b) Key Issues: The random black/white specks are visually similar to text characters (dots, punctuation), which directly interferes with OCR accuracy and human readability. The noise is high-frequency and high-contrast, so it can be mistaken for genuine document features such as periods, commas, or fine strokes. It also corrupts thin character strokes, making letters appear broken or merged.

(c) Applied Technique: A median filter is applied. A square neighbourhood window (e.g., 3x3) is slid across the image; each pixel is replaced by the median (not the mean) of the intensity values in that window, which discards the extreme outlier values introduced by the noise.

(d) Justification: Salt-and-pepper noise consists of extreme outliers, so an order-statistics filter is the natural fit — the median is inherently robust to outliers and simply excludes them from the output. A mean/averaging or Gaussian filter would instead blend the 0/255 outlier into the neighbourhood average, spreading the corruption and blurring text edges rather than removing it. The median filter therefore removes the noise while keeping character edges and stroke boundaries sharp, which is critical for readability and OCR.

(e) Presentation: Summary: Noise type = impulsive (salt-and-pepper) -> Problem = broken/misread text -> Chosen method = median filtering -> Outcome = noise removed with edges preserved, restoring document readability.

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2. Object Boundary Detection (Medical Image)

(a) Scenario & Importance: In medical imaging (e.g., MRI/CT/ultrasound), tissues of different types often have similar intensity ranges with subtle transitions between them. Accurately locating the boundary between tissues (e.g., tumour vs. healthy tissue, organ vs. background) is essential for correct diagnosis, treatment planning, volume/area measurement, and downstream tasks such as segmentation-based quantification.

(b) Key Challenges: Medical images often have low contrast between adjacent tissues, inherent acquisition noise (speckle in ultrasound, thermal noise in MRI), intensity inhomogeneity/uneven illumination, and blurred or gradual transitions rather than sharp intensity jumps — all of which make simple gradient-based detection prone to false or missed edges.

(c) Applied Technique: The Canny edge detector is applied. It proceeds in stages: Gaussian smoothing to suppress noise, gradient magnitude and direction computation (Sobel), non-maximum suppression to thin edges to a single-pixel width, and double (hysteresis) thresholding with edge tracking to link weak edges connected to strong ones while discarding isolated noise responses.

(d) Justification: Canny is preferred over simpler operators (Sobel, Prewitt, Roberts) because it explicitly balances noise suppression with edge localisation and produces thin, well-connected, single-pixel-wide boundaries rather than thick or scattered edge fragments. Its hysteresis thresholding is particularly valuable in medical images, where a true tissue boundary may have a weak gradient in places — Canny can still retain it if it connects to a strong-gradient segment, whereas a single global threshold would break the boundary.

(e) Presentation: Summary: Requirement = precise tissue boundary -> Challenge = low contrast/noise/gradual transitions -> Chosen method = Canny edge detection -> Outcome = thin, continuous, noise-robust boundaries suitable for clinical interpretation.

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3. Simple Segmentation (Intensity-Separable Object)

(a) Scenario & Requirement: When an object is clearly distinguishable from its background based on intensity — e.g., a dark object on a light background — the image histogram is typically bimodal, with one peak for the object and one for the background, separated by a valley. Segmentation here only needs to classify each pixel as object or background using intensity, without complex boundary tracing.

(b) Key Enabling Features: The bimodal histogram itself is the key feature: the two well-separated intensity clusters and the low-density valley between them indicate that a single intensity value can cleanly split the image into two classes with minimal misclassification.

(c) Applied Technique: Global thresholding using Otsu's method is applied. Otsu automatically searches over all possible threshold values and selects the one that maximises the between-class variance (equivalently minimises within-class variance) of the resulting foreground/background pixel groups, then produces a binary image.

(d) Justification: Because the histogram is bimodal, a single global threshold is sufficient and computationally cheap — there is no need for adaptive/local thresholding or more complex region-growing or clustering methods, which would add unnecessary computation for no gain in accuracy. Otsu's method is preferred over a manually chosen threshold because it is derived directly and objectively from the image's own statistics, making it repeatable and not dependent on subjective tuning.

(e) Presentation: Summary: Condition = bimodal intensity separation -> Feature used = histogram peaks/valley -> Chosen method = Otsu global thresholding -> Outcome = clean binary segmentation of object from background.

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4. Low Brightness Image

(a) Scenario & Cause: An image captured under low-light conditions has most of its pixel intensities compressed into the lower (dark) end of the 0-255 range. This results from insufficient scene illumination, a short exposure time/small aperture, or low sensor sensitivity (ISO), all of which limit the amount of light reaching the sensor and thus the recorded intensity values.

(b) Key Issues: The narrow, low-valued histogram means poor visibility of details, low overall contrast, and difficulty distinguishing objects from shadows or from each other. Because the dynamic range is compressed, information is technically present but visually indistinguishable to the human eye.

(c) Applied Technique: Histogram equalisation is applied (adaptive/CLAHE if regional variation is high) to redistribute the pixel intensities so that they span the full available range, increasing global contrast.

(d) Justification: Simply adding a constant brightness offset would shift all values up uniformly but risks clipping/saturating already-bright pixels and does not necessarily improve contrast between similar dark tones. Histogram equalisation instead stretches the cumulative distribution of intensities so darker regions are spread across a wider range, revealing detail that was previously compressed into a few intensity levels. CLAHE further limits noise amplification in near-uniform dark regions by operating on local tiles with contrast limiting, which plain global equalisation can over-amplify.

(e) Presentation: Summary: Cause = insufficient light/exposure -> Problem = compressed, low-value histogram -> Chosen method = histogram equalisation (CLAHE for local detail) -> Outcome = improved visibility and contrast without excessive noise amplification.

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5. Gaussian Noise (Smooth Regions)

(a) Scenario & Description: Gaussian noise is additive noise where the intensity added to each pixel is drawn from a normal (Gaussian) distribution with zero mean and some variance. It commonly arises from sensor thermal noise and electronic circuit noise during image acquisition. Unlike salt-and-pepper noise, it affects every pixel by a small, continuous random amount rather than a few pixels by an extreme amount.

(b) Key Issues: Smooth, uniform regions (sky, skin, flat backgrounds) that should have near-constant intensity instead appear grainy/speckled, reducing perceived image quality and lowering the signal-to-noise ratio, which can also interfere with later processing such as edge detection or segmentation.

(c) Applied Technique: A Gaussian smoothing (low-pass) filter is applied, convolving the image with a Gaussian kernel that performs a weighted local average, giving more weight to nearby pixels.

(d) Justification: Since the noise itself follows a Gaussian statistical model, a Gaussian filter is a matched, principled choice for averaging it out while still weighting nearby true-signal pixels more heavily than distant ones, producing a smoother result than a plain box/mean filter of the same size. A median filter is less suitable here because Gaussian noise is not impulsive — every pixel carries a small error, so an outlier-rejecting filter offers little benefit and can needlessly discard genuine fine detail.

(e) Presentation: Summary: Noise type = additive Gaussian -> Problem = graininess in smooth regions -> Chosen method = Gaussian smoothing filter -> Outcome = reduced noise variance with acceptable, controlled loss of fine detail.

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6. Loss of Fine Details After Processing

(a) Scenario & Explanation: After noise removal or smoothing, an image can lose fine details such as sharp edges, thin lines, and texture. This happens because low-pass/smoothing filters (mean, Gaussian, or aggressive median filtering with a large window) do not distinguish between noise and genuine high-frequency image content — both are attenuated together.

(b) Key Issues: There is an inherent trade-off between noise suppression and detail preservation: a filter strong enough to remove noise effectively also blurs edges and textures, reducing sharpness and making the image look soft or out of focus.

(c) Applied Technique: Unsharp masking (high-boost filtering) is applied: a blurred version of the image is subtracted from the original to isolate high-frequency detail, and this detail is then added back to the original with a weighting factor to boost sharpness. Where noise is still a concern, an edge-preserving smoothing filter (e.g., bilateral filter) can be used upstream instead of a plain Gaussian filter.

(d) Justification: Unsharp masking directly targets the high-frequency components that were attenuated, re-emphasising edges and fine structures without having to redo the whole processing pipeline. Using an edge-preserving filter such as the bilateral filter earlier in the pipeline is justified because it smooths based on both spatial distance and intensity similarity, so it averages within homogeneous regions but avoids averaging across edges — unlike a plain Gaussian filter, which blurs indiscriminately regardless of edge content.

(e) Presentation: Summary: Cause = indiscriminate low-pass filtering -> Problem = blurred edges/lost texture -> Chosen method = unsharp masking (and/or bilateral filtering) -> Outcome = restored sharpness/detail while retaining noise control.

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7. Edge Enhancement Requirement (Feature Extraction)

(a) Scenario & Need: An application intends to extract features (shapes, corners, object outlines) for tasks such as recognition or classification, and therefore needs edges — the locations of rapid intensity change — to be highlighted clearly so that subsequent feature-extraction stages can operate on well-defined boundaries.

(b) Key Issues: Real images contain both genuine edges and noise, and naive differentiation-based operators amplify noise as much as true edges. Weak but real edges (e.g., low-contrast object boundaries) may be difficult to distinguish from noise-induced gradients, and directionality of edges (horizontal, vertical, diagonal) must also be captured for reliable feature descriptors.

(c) Applied Technique: The Sobel operator is applied, convolving the image with two 3x3 kernels (horizontal and vertical) to estimate the intensity gradient in each direction; the gradient magnitude and direction are then combined to highlight edges. A Laplacian-of-Gaussian (LoG) can be used as an alternative when a rotation-invariant, second-derivative response is preferred.

(d) Justification: Sobel is favoured for general edge-enhancement/feature-extraction pipelines because it combines a smoothing effect (via its kernel weighting) with differentiation in a single pass, making it less sensitive to noise than a raw first-derivative (Roberts) operator, while still being computationally light enough for real-time feature pipelines. It also yields both gradient magnitude (edge strength) and direction, which downstream feature descriptors (e.g., HOG-style features) can use directly, something a simple thresholding-based method cannot provide.

(e) Presentation: Summary: Requirement = highlighted edges for features -> Challenge = noise vs. true edges, directionality -> Chosen method = Sobel operator (LoG as alternative) -> Outcome = enhanced, direction-aware edge map ready for feature extraction.
